

compensate for the speaker. If it compensates often, students may never discover that they pronounced a word badly; if it compensates too little, there would be too many errors.

The application of SR towards helping people with disabilities such as dyslexia or dysgraphia, is yet another challenge. Those who cannot use a keyboard, or notepad, can benefit tremendously from SR technology. However, the software would need to be tailored to individual cases, because of variations of the conditions amongst sufferers.

Among the many commercial applications of SR, is the automation of call centres and the like, as well as voice systems that take orders over the phone. A problem area here is the variability in telephone microphones, along with, of course, a caller's telephone habits—such as how far away from the mouth he or she holds the mouthpiece. Such difficulties are not insurmountable, but they go to show that research that merely attempts to improve accuracy is not enough.

What's being done

Although SR has been around for quite a while, it's only recently that it's taken off. One reason is that processor and memory requirements are relatively high, and it's only now, when memories and processors are more powerful and affordable, that powerful SR applications have been made possible.

Research into good models of how we construct our speech will continue. SR technology may benefit from research in

Areas Of Application

Flight and combat: The British Eurofighter Typhoon is the first combat aircraft to have speech recognition as a standard part of the input system. The pilot can issue voice instructions, and have spoken questions answered by the system. This way, the pilot need not look down at buttons, dials, LCD screens, etc., and is free to navigate in a more natural manner.

Automated surgery: AESOP 2000 is a surgical robot that takes commands that tell it where to hold a particular instrument. This is used in keyhole surgery, which involves the surgeon looking through a tiny opening in the skin, as happens in endoscopy.

Translation: Automated translation has

been researched for long, and translation systems for PDAs will soon hit the market. The speaker speaks in his native language into the PDA, which then translates the text into the chosen language, and speaks it aloud.

Household appliances: Sensory Incorporated sells their VoiceDirect Speech Recognition Kit for less than \$50 (approximately Rs 2,300). The kit comes with everything one needs to speech-enable household appliances and devices such as remotes and telephones. One simply needs to wire up the devices according to the setup procedure, and train the system with commands, such as 'power on'.

other areas—Natural Language Processing, and Machine Learning, for example. The phonemic model of speech has been very successful, but it is by no means the only model. At CMU, alternatives such as the modelling of speech based on syllables are being studied. Also being developed at CMU, are language models that automatically detect, and adapt to, changes in the form and the content of what is being spoken—the speaker's pronunciation and vocabulary, and even what he, or she, is talking about.

Yet another problem being addressed at CMU is that of the transcribing of pronunciations for thousands and thousands of words: in the training phase of an SR system, a linguist needs to read out volumes of text, in a consistent tone, volume, etc. Methods of automatically generating pronunciations for new words, from training examples, are being devised.

What we should see

Speech and handwriting recognition technologies can only get better each year. The fact that Microsoft bundles dictation and handwriting software with Office XP 2003, is an indication of how ubiquitous they will become.

It is expected that pen input systems will replace some, if not all, keyboards. Devices such as the SmartQuill hold special promise.

Pen computing may have other implications as well. The Roman alphabet has a special status in the world today. More ubiquitous speech and handwriting recognition could change that. As things stand, people who don't use the Roman alphabet are on the other side of the 'keyboard

divide': using computers entails acquiring keyboard skills that don't come intuitively. Consider someone who speaks an Indian language, he or she needs to press three keys on a QWERTY keyboard to get a character, and think of what pen-based input could do for him or her.

Also, it's natural that translation services, together with natural-input recognition systems, would bridge international computing boundaries.

Peter Fleming, president of Aristotle Systems, a speech-technology consulting firm, has some realistic predictions for SR in the years to come. He predicts, "Greater 'programmed understanding' of grammar and syntax will improve accurate recognition... Keyboards, mice and electronic pens will become more closely integrated with speech recognition... Microphone positioning and usage will become more natural; more use will be made of common sense in choosing the correct word for a particular context." However, he also says, "Dictation speech recognition will never become the universal and exclusive data input method."

As we inch towards transparent computing—and eventually Ubicomp—one small step at a time, two things are clear: advances in speech and handwriting recognition will drive much of the transition; and, as we learn to communicate more directly with our machines, there must be a compromise. Until NLP and other AI problems are solved, we will continue to compensate for what the machine lacks. And hope that, as time advances, this needed compensation will diminish. ■

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www.technologyreview.com/articles/rmb_121903.asp:

A system whereby a speaker speaks into a PDA in one language, and the PDA speaks out a translation in another

www.india-today.com/ctoday/19991201/marvels.html:

A description of how speech recognition is incorporated in the Eurofighter Typhoon

<http://research.microsoft.com/news/msrnews/smartquill.aspx>:

A page about the SmartQuill, a device that captures handwriting strokes in its own memory

www.pencomputing.com: A perspective on pen computing, current and future

[www.ubiq.com/hypertext/weiser/](http://www.ubiq.com/hypertext/weiser/UbiHome.html)

[UbiHome.html](http://www.ubiq.com/hypertext/weiser/UbiHome.html): Mark Weiser's page on Ubiquitous Computing, with several write-ups and papers